

CHAPTER C30

WIND LOADS: COMPONENTS AND CLADDING

In developing the set of pressure coefficients applicable for the design of components and cladding (C&C) as given in Figs. 30.3-1, 30.3-2A–C, 30.3-3, 30.3-4, 30.3-5A–B, and 30.3-6, an envelope approach was followed but using different methods than for the main wind-force resisting system (MWFRS) of Fig. 28.3-1. Because of the small effective area that may be involved in the design of a particular component (consider, for example, the effective area associated with the design of a fastener), the pointwise pressure fluctuations may be highly correlated over the effective area of interest.

Consider the local purlin loads shown in Fig. C28.3-1. The approach involved spatial averaging and time averaging of the point pressures over the effective area transmitting loads to the purlin while the building model was permitted to rotate in the wind tunnel through 360°. As the induced localized pressures may also vary widely as a function of the specific location on the building, height above ground level, exposure, and, more importantly, local geometric discontinuities and location of the element relative to the boundaries in the building surfaces (e.g., walls, roof lines), these factors were also enveloped in the wind tunnel tests. Thus, for the pressure coefficients given in Figs. 30.3-1, 30.3-2A–C, 30.3-3, 30.3-4, 30.3-5A–B, and 30.3-6, the directionality of the wind and influence of exposure have been removed and the surfaces of the building have been “zoned” to reflect an envelope of the peak pressures possible for a given design application.

For ASCE 7-16, the roof zones and pressure coefficients for Fig. 30.3-2A were modified based on the analysis by Kopp and Morrison (2014), which made use of the extensive wind tunnel database developed by Ho et al. (2005). St. Pierre et al. (2005) provided an evaluation of this database compared to earlier data by Davenport et al. (1977, 1978) and ASCE 7 (2002), while Ho et al. (2005) compared the data to full-scale field data from Texas Tech University (Mehta and Levitan 1998). All source data used in the study are publicly accessible through the National Institute of Standards and Technology’s website (see, e.g., Main and Fritz 2006). Compared to previous versions of ASCE 7, the pressure coefficients have been increased and are now more consistent with coefficients for buildings higher than 60 ft (18.3 m). Roof zone sizes are also modified from those of earlier versions in order to minimize the increase of pressure coefficients in Zones 1 and 2. The data indicate that for these low-rise buildings, the size of the roof zones depends primarily on the building height, h . A Zone 1 now occurs for large buildings, which accounts for the lower wind loads in the middle of the roof. Zone 3 (roof corner) is an “L” shape, consistent with the shape of Zone 3 for buildings higher than 60 ft (18.3 m) and consistent with the wind loading data. Four potential zone configurations based on the ratios of the smallest and largest building plan dimensions are illustrated in Fig. C30-1. In addition, when the greatest horizontal dimension

is less than $0.4h$ (the building does not correspond to a typical low-rise building shape), there is a single roof zone (Zone 3). Detailed explanations can be found in Kopp and Morrison (2014).

As indicated in the discussion for Fig. 28.3-1, the wind tunnel experiments checked both Exposure B and C terrains. Basically, (GC_p) values associated with Exposure B terrain would be higher than those for Exposure C terrain because of reduced velocity pressure in Exposure B terrain. The (GC_p) values given in Figs. 30.3-1, 30.3-2A–C, 30.3-3, 30.3-4, 30.3-5A–B, and 30.3-6 are associated with Exposure C terrain as obtained in the wind tunnel. However, they may also be used for any exposure when the correct velocity pressure representing the appropriate exposure is used. The (GC_p) values given in Figs. 30.3-2A–C are associated with wind tunnel tests performed in both Exposures B and C.

For Fig. 30.3-2A, the coefficients apply equally to Exposure B and C, based on wind tunnel data that show insignificant differences in (GC_p) for Exposures B and C. Consequently, the truncation for K_z in Table 30.3-1 of ASCE 7-10 is not required for buildings below 30 ft (9.1 m), and the lower K_z values may be used.

The pressure coefficients given in Fig. 30.5-1 for buildings with mean height greater than 60 ft (18.3 m) were developed following a similar approach, but the influence of exposure was not enveloped (Stathopoulos and Dumitrescu-Brulotte 1989). Therefore, exposure categories B, C, or D may be used with the values of (GC_p) in Fig. 30.5-1 as appropriate.

C30.1 SCOPE

C30.1.1 Building Types. Guidance for determining C_f and A_f for C&C of structures found in petrochemical and other industrial facilities that are not otherwise addressed in ASCE 7 can be found in *Wind Loads for Petrochemical and Other Industrial Facilities* (ASCE 2011). The 2011 edition references ASCE 7-05, and the user needs to make appropriate adjustments where compliance with the ASCE 7-10 standard is required.

C30.1.5 Air-Permeable Cladding. Air-permeable roof or wall claddings allow partial air pressure equalization between their exterior and interior surfaces. Examples include siding, pressure-equalized rain screen walls, shingles, tiles (including modular vegetative roof assemblies), concrete roof pavers, and aggregate roof surfacing.

The peak pressure acting across an air-permeable cladding material is dependent on the characteristics of other components or layers of a building envelope assembly. At any given instant, the total net pressure across a building envelope assembly is equal to the sum of the partial pressures across the individual layers, as shown in Fig. C30.1-1. However, the proportion of the total net pressure borne by each layer varies from instant to

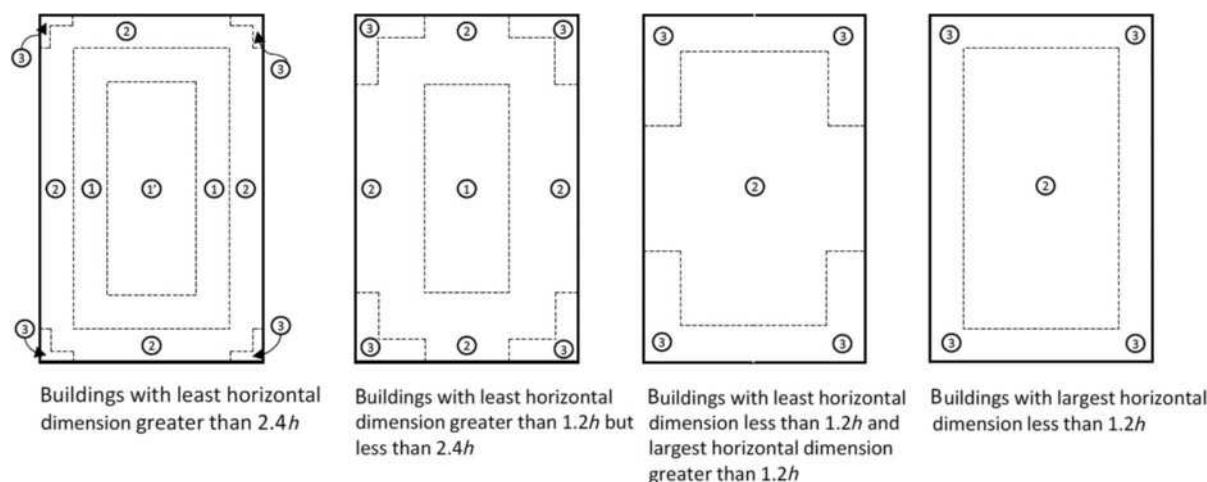


FIGURE C30-1 Four Possible Scenarios for Roof Zones, Which Depend on the Ratios of the Least and Largest Horizontal Plan Dimensions to the Mean Roof Height h

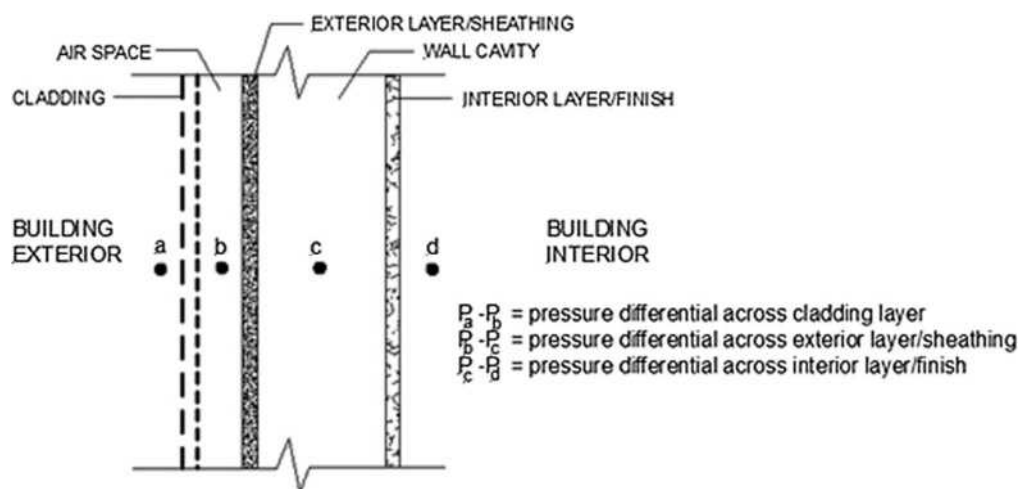


FIGURE C30.1-1 Distribution of Net Components and Cladding Pressure Acting on a Building Surface (Building Envelope) Composed of Three Components (Layers)

instant because of fluctuations in the external and internal pressures and depends on the porosity and stiffness of each layer, as well as the volumes of the air spaces between the layers. As a result, although there is load sharing among the various layers, the sum of the peak pressures across the individual layers typically exceeds the peak pressure across the entire system. In the absence of detailed information on the division of loads, a simple, conservative approach is to assign the entire differential pressure to each layer designed to carry load.

To maximize pressure equalization (reduction) across any cladding system (irrespective of the permeability of the cladding itself), the layer or layers behind the cladding should be

- Relatively stiff in comparison to the cladding material; and
- Relatively air-impermeable in comparison to the cladding material.

Furthermore, the air space between the cladding and the next adjacent building envelope surface behind the cladding (e.g., the exterior sheathing) should be as small as practicable and compartmentalized to avoid communication or venting between different pressure zones of a building's surfaces.

The design wind pressures derived from Chapter 30 represent the pressure differential between the exterior and interior surfaces of the exterior envelope (wall or roof system). Because of partial air-pressure equalization provided by air-permeable claddings, the C&C pressures derived from Chapter 30 can overestimate the load on air-permeable cladding elements. The designer may elect either to use the loads derived from Chapter 30 or to use loads derived by an approved alternative method. If the designer desires to determine the pressure differential across a specific cladding element in combination with other elements comprising a specific building envelope assembly, appropriate pressure measurements should be made on the applicable building envelope assembly, or reference should be made to recognized literature (Cheung and Melbourne 1986; Haig 1990; Baskaran 1992; SBCCI 1994; Peterka et al. 1997; ASTM 2006, 2007; Kala et al. 2008; Baskaran et al. 2012; Kopp and Gavanski 2012; and Cope et al. 2012) for documentation pertaining to wind loads. Such alternative methods may vary according to a given cladding product or class of cladding products or assemblies because each has unique features that affect pressure equalization. It is important to consider the methodology used to determine wind

pressure distribution through a multilayered assembly including an air-permeable cladding layer. Recent full-scale wind tunnel tests have shown that an accurate distribution of the wind pressure in a multilayered exterior wall assembly must account for the spatial and temporal (dynamic) fluctuations in wind pressure representative of actual wind flow conditions (Cope et al. 2012). Other factors to consider include the influence of air-flow pathways through the assembly (e.g., openings or penetrations through any given layer) and appropriate methods of enveloping peak pressure coefficients for each layer of a multilayered assembly (e.g., Cope et al. 2012) to ensure system reliability and consistency with the characterization of peak pressure coefficients in this standard.

Modular Vegetative Roof Assemblies consist of vegetation and other components integrated as a tray. These trays have vertical air gaps (a minimum of 0.25 in. (6.25 mm)) between the module and roofing system and horizontal air gaps between them. These air gaps allow partial air pressure equalization.

C30.3 BUILDING TYPES

C30.3.1 Conditions. For velocity pressure, see commentary, Section C26.10.1.

C30.3.2 Design Wind Pressures. For velocity pressure, see commentary, Section C26.10.1.

Figs. 30.3-1 and 30.3-2A–C. The pressure coefficient values provided in these figures are to be used for buildings with a mean roof height of 60 ft (18.3 m) or less. The values were obtained from wind tunnel tests conducted at the University of Western Ontario (Davenport et al. 1977, 1978; Ho et al. 2005; St. Pierre et al. 2005; Kopp and Morrison 2014; Vickery et al. 2011; Gavanski et al. 2013). The negative roof (GC_p) values given in these figures are significantly greater (in magnitude) than those given in previous versions (2010 and earlier) but are consistent with those given in Ho et al. (2005). The (GC_p) values given in the figures are given in equation form in Tables C30.3-1 to C30.3-10. Note that the (GC_p) values given in Fig. 30.3-2A–C are a function of the roof slope. Some of the characteristics of the values in the figure are as follows:

1. The values are combined values of (GC_p). The gust-effect factors from these values should not be separated;
2. The velocity pressure, q_h , evaluated at mean roof height should be used with all values of (GC_p);
3. The values provided in the figure represent the upper bounds of the most severe values for any wind direction. The reduced probability that the design wind speed may not occur in the particular direction for which the worst pressure coefficient is recorded has not been included in the values shown in the figure; and
4. The wind tunnel values, as measured, were based on the mean hourly wind speed. The values provided in the figures

Table C30.3-1. Walls for Buildings with $h \leq 60$ ft ($h \leq 18.3$ m) (Figure 30.3-1)

Positive:	$(GC_p) = 1.0$	for $A = 10 \text{ ft}^2$
Zones 4 and 5	$(GC_p) = 1.1766 - 0.1766 \log A$	for $10 < A \leq 500 \text{ ft}^2$
	$(GC_p) = 0.7$	for $A > 500 \text{ ft}^2$
Negative:	$(GC_p) = -1.1$	for $A = 10 \text{ ft}^2$
Zone 4	$(GC_p) = -1.2766 + 0.1766 \log A$	for $10 < A \leq 500 \text{ ft}^2$
	$(GC_p) = -0.8$	for $A > 500 \text{ ft}^2$
Negative:	$(GC_p) = -1.4$	for $A = 10 \text{ ft}^2$
Zone 5	$(GC_p) = -1.7532 + 0.3532 \log A$	for $10 < A \leq 500 \text{ ft}^2$
	$(GC_p) = -0.8$	for $A > 500 \text{ ft}^2$

are the measured values divided by the 3-second dynamic gust pressure at mean roof height to adjust for the reduced pressure coefficient values associated with a 3-s gust speed.

Table C30.3-2. Gable Roof, $\theta \leq 7^\circ$ (Figure 30.3-2A)

Positive with and without overhang		
All Zones	$(GC_p) = 0.3$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = 0.4000 - 0.1000 \log A$	for $10 \leq A \leq 100 \text{ ft}^2$
	$(GC_p) = 0.2$	for $A \geq 100 \text{ ft}^2$
Negative without overhang		
Zone 1'	$(GC_p) = -0.9$	for $A \leq 100 \text{ ft}^2$
	$(GC_p) = -1.9000 + 0.5000 \log A$	for $100 \leq A \leq 1,000 \text{ ft}^2$
	$(GC_p) = -0.4$	for $A \geq 1,000 \text{ ft}^2$
Zone 1	$(GC_p) = -1.7$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -2.1120 + 0.4120 \log A$	for $10 \leq A \leq 500 \text{ ft}^2$
	$(GC_p) = -1.0$	for $A \geq 500 \text{ ft}^2$
Zone 2	$(GC_p) = -2.3$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -2.8297 + 0.5297 \log A$	for $10 \leq A \leq 500 \text{ ft}^2$
	$(GC_p) = -1.4$	for $A \geq 500 \text{ ft}^2$
Zone 3	$(GC_p) = -3.2$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -4.2595 + 1.0595 \log A$	for $10 \leq A \leq 500 \text{ ft}^2$
	$(GC_p) = -1.4$	for $A \geq 500 \text{ ft}^2$
Negative with overhang		
Zone 1 and 1'	$(GC_p) = -1.7$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -1.8000 + 0.1000 \log A$	for $10 \leq A \leq 100 \text{ ft}^2$
	$(GC_p) = -3.3168 + 0.8584 \log A$	for $100 \leq A \leq 500 \text{ ft}^2$
	$(GC_p) = -1.0$	for $A \geq 500 \text{ ft}^2$
Zones 2	$(GC_p) = -2.3$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -3.0063 + 0.7063 \log A$	for $10 \leq A \leq 500 \text{ ft}^2$
	$(GC_p) = -1.1$	for $A \geq 500 \text{ ft}^2$
Zone 3	$(GC_p) = -3.2$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -4.4360 + 1.2360 \log A$	for $10 \leq A \leq 500 \text{ ft}^2$
	$(GC_p) = -1.1$	for $A \geq 500 \text{ ft}^2$

Table C30.3-3. Gable Roof, $7^\circ < \theta \leq 20^\circ$ (Figure 30.3-2B)

Positive with and without overhang		
All Zones	$(GC_p) = 0.7$	for $A \leq 2 \text{ ft}^2$
	$(GC_p) = 0.7709 - 0.2354 \log A$	for $2 \leq A \leq 100 \text{ ft}^2$
	$(GC_p) = 0.3$	for $A \geq 100 \text{ ft}^2$
Negative without overhang		
Zones 1 and 2e	$(GC_p) = -2.0$	for $A \leq 20 \text{ ft}^2$
	$(GC_p) = -4.7920 + 2.1460 \log A$	for $20 \leq A \leq 100 \text{ ft}^2$
	$(GC_p) = -0.5$	for $A \geq 100 \text{ ft}^2$
Zones 2n, 2r, and 3e	$(GC_p) = -3.0$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -4.4307 + 1.4307 \log A$	for $10 \leq A \leq 250 \text{ ft}^2$
	$(GC_p) = -1.0$	for $A \geq 250 \text{ ft}^2$
Zone 3r	$(GC_p) = -3.6$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -5.4000 + 1.800 \log A$	for $10 \leq A \leq 100 \text{ ft}^2$
	$(GC_p) = -1.8000$	for $A \geq 100 \text{ ft}^2$
Negative with overhang		
Zones 1 and 2e	$(GC_p) = -2.5$	for $A \leq 20 \text{ ft}^2$
	$(GC_p) = -4.3614 + 1.4307 \log A$	for $20 \leq A \leq 100 \text{ ft}^2$
	$(GC_p) = -1.5$	for $A \geq 100 \text{ ft}^2$
Zones 2n and 2r	$(GC_p) = -3.5$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -4.5730 + 1.0730 \log A$	for $10 \leq A \leq 250 \text{ ft}^2$
	$(GC_p) = -2.0$	for $A \geq 250 \text{ ft}^2$
Zone 3e	$(GC_p) = -4.1$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -5.9599 + 1.8599 \log A$	for $10 \leq A \leq 250 \text{ ft}^2$
	$(GC_p) = -1.5$	for $A \geq 250 \text{ ft}^2$
Zone 3r	$(GC_p) = -4.7$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -7.1000 + 2.4000 \log A$	for $10 \leq A \leq 100 \text{ ft}^2$
	$(GC_p) = -2.3$	for $A \geq 100 \text{ ft}^2$

Table C30.3-4. Gable Roofs, $20^\circ < \theta \leq 27^\circ$ (Figure 30.3-2C)

Positive with and without overhang		
All Zones	$(GC_p) = 0.7$	for $A \leq 2 \text{ ft}^2$
	$(GC_p) = 0.7709 - 0.2354 \log A$	for $2 \leq A \leq 100 \text{ ft}^2$
	$(GC_p) = 0.3$	for $A \geq 100 \text{ ft}^2$
Negative without overhang		
Zones 1 and 2e	$(GC_p) = -1.5$	for $A \leq 20 \text{ ft}^2$
	$(GC_p) = -2.2744 + 0.5952 \log A$	for $20 \leq A \leq 300 \text{ ft}^2$
	$(GC_p) = -0.8$	for $A \geq 300 \text{ ft}^2$
Zones 2n, 2r, and 3r	$(GC_p) = -2.5$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -3.6054 + 1.1054 \log A$	for $10 \leq A \leq 150 \text{ ft}^2$
	$(GC_p) = -1.2$	for $A \geq 150 \text{ ft}^2$
Zones 3e	$(GC_p) = -3.6$	for $A \leq 4 \text{ ft}^2$
	$(GC_p) = -4.5880 + 1.6410 \log A$	for $4 \leq A \leq 50 \text{ ft}^2$
	$(GC_p) = -1.8$	for $A \geq 50 \text{ ft}^2$
Negative with overhang		
Zones 1 and 2e	$(GC_p) = -2.0$	for $A \leq 20 \text{ ft}^2$
	$(GC_p) = -2.2212 + 0.1701 \log A$	for $20 \leq A \leq 300 \text{ ft}^2$
	$(GC_p) = -1.8$	for $A \geq 300 \text{ ft}^2$
Zones 2n and 2r	$(GC_p) = -3.0$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -3.6802 + 0.6802 \log A$	for $10 \leq A \leq 150 \text{ ft}^2$
	$(GC_p) = -2.2$	for $A \geq 150 \text{ ft}^2$
Zone 3r	$(GC_p) = -3.6$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -5.2155 + 1.6155 \log A$	for $10 \leq A \leq 150 \text{ ft}^2$
	$(GC_p) = -1.7$	for $A \geq 150 \text{ ft}^2$
Zone 3e	$(GC_p) = -4.7$	for $A \leq 4 \text{ ft}^2$
	$(GC_p) = -6.0173 + 2.1880 \log A$	for $4 \leq A \leq 50 \text{ ft}^2$
	$(GC_p) = -2.3$	for $A \geq 50 \text{ ft}^2$

Table C30.3-5. Gable Roofs, $27^\circ < \theta \leq 45^\circ$ (Figure 30.3-2D)

Positive with and without overhang		
All Zones	$(GC_p) = 0.9$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = 1.3000 - 0.4000 \log A$	for $10 \leq A \leq 100 \text{ ft}^2$
	$(GC_p) = 0.5$	for $A \geq 100 \text{ ft}^2$
Negative without overhang		
Zones 1, 2e, and 2r	$(GC_p) = -1.8$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -2.8000 + 1.0000 \log A$	for $10 \leq A \leq 100 \text{ ft}^2$
	$(GC_p) = -0.8$	for $A \geq 100 \text{ ft}^2$
Zones 2n and 3r	$(GC_p) = -2.0$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -2.7686 + 0.7686 \log A$	for $10 \leq A \leq 200 \text{ ft}^2$
	$(GC_p) = -1.0$	for $A \geq 200 \text{ ft}^2$
Zone 3e	$(GC_p) = -3.2$	for $A \leq 2 \text{ ft}^2$
	$(GC_p) = -3.5043 + 1.0110 \log A$	for $2 \leq A \leq 300 \text{ ft}^2$
	$(GC_p) = -1.0$	for $A \geq 300 \text{ ft}^2$
Negative with overhang		
Zones 1, 2e, and 2r	$(GC_p) = -2.6$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -3.6000 + 1.0000 \log A$	for $10 \leq A \leq 100 \text{ ft}^2$
	$(GC_p) = -1.6$	for $A \geq 100 \text{ ft}^2$
Zones 2n and 3r	$(GC_p) = -2.8$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -3.5686 + 0.7686 \log A$	for $10 \leq A \leq 200 \text{ ft}^2$
	$(GC_p) = -1.8$	for $A \geq 200 \text{ ft}^2$
Zone 3e	$(GC_p) = -4.0$	for $A \leq 2 \text{ ft}^2$
	$(GC_p) = -4.3043 + 1.0110 \log A$	for $2 \leq A \leq 300 \text{ ft}^2$
	$(GC_p) = -1.8$	for $A \geq 300 \text{ ft}^2$

Each C&C element should be designed for the maximum positive and negative pressures (including applicable internal pressures) acting on it. The pressure coefficient values should be determined for each C&C element on the basis of its location on the building and the effective area for the element. Research (Stathopoulos and Zhu 1988, 1990) indicated that the pressure

Table C30.3-6. Hip Roofs, No Overhang, $7^\circ < \theta \leq 20^\circ$ (Figure 30.3-2E)

Positive $h/B \geq 0.8$		
All Zones	$(GC_p) = 0.7$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = 1.1000 - 0.4000 \log A$	for $10 \leq A \leq 100 \text{ ft}^2$
	$(GC_p) = 0.3$	for $A \geq 100 \text{ ft}^2$
Negative $h/B \geq 0.8$		
Zone 1	$(GC_p) = -1.8$	for $A \leq 20 \text{ ft}^2$
	$(GC_p) = -3.2891 + 1.1445 \log A$	for $20 \leq A \leq 100 \text{ ft}^2$
	$(GC_p) = -1.0$	for $A \geq 100 \text{ ft}^2$
Zone 2r	$(GC_p) = -2.4$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -3.2455 + 0.8455 \log A$	for $10 \leq A \leq 200 \text{ ft}^2$
	$(GC_p) = -1.3$	for $A \geq 200 \text{ ft}^2$
Zones 2e and 3	$(GC_p) = -2.6$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -3.5223 + 0.9223 \log A$	for $10 \leq A \leq 200 \text{ ft}^2$
	$(GC_p) = -1.4$	for $A \geq 200 \text{ ft}^2$
Positive $h/B \leq 0.5$		
All Zones	$(GC_p) = 0.7$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = 1.1000 - 0.4000 \log A$	for $10 \leq A \leq 100 \text{ ft}^2$
	$(GC_p) = 0.3$	for $A \geq 100 \text{ ft}^2$
Negative $h/B \leq 0.5$		
Zone 1	$(GC_p) = -1.3$	for $A \leq 20 \text{ ft}^2$
	$(GC_p) = -1.8584 + 0.4292 \log A$	for $20 \leq A \leq 100 \text{ ft}^2$
	$(GC_p) = -1.0$	for $A \geq 100 \text{ ft}^2$
Zone 2r	$(GC_p) = -2.4$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -3.2455 + 0.8455 \log A$	for $10 \leq A \leq 200 \text{ ft}^2$
	$(GC_p) = -1.3$	for $A \geq 200 \text{ ft}^2$
Zones 2e and 3	$(GC_p) = -1.8$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -2.3380 + 0.5380 \log A$	for $10 \leq A \leq 200 \text{ ft}^2$
	$(GC_p) = -1.1$	for $A \geq 200 \text{ ft}^2$

Table C30.3-7. Hip Roofs, Overhang, $7^\circ < \theta \leq 20^\circ$ (Figure 30.3-2F)

Negative $h/B \geq 0.8$		
Zone 1	$(GC_p) = -2.3$	for $A \leq 20 \text{ ft}^2$
	$(GC_p) = -2.8584 + 0.4292 \log A$	for $20 \leq A \leq 100 \text{ ft}^2$
	$(GC_p) = -2.0$	for $A \geq 100 \text{ ft}^2$
Zone 2r	$(GC_p) = -2.9$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -3.3612 + 0.4612 \log A$	for $10 \leq A \leq 200 \text{ ft}^2$
	$(GC_p) = -2.3$	for $A \geq 200 \text{ ft}^2$
Zones 2e	$(GC_p) = -3.1$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -3.6380 + 0.5380 \log A$	for $10 \leq A \leq 200 \text{ ft}^2$
	$(GC_p) = -2.4$	for $A \geq 200 \text{ ft}^2$
Zones 3	$(GC_p) = -3.7$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -5.0835 + 1.3835 \log A$	for $10 \leq A \leq 200 \text{ ft}^2$
	$(GC_p) = -1.9$	for $A \geq 200 \text{ ft}^2$
Negative $h/B \leq 0.5$		
Zone 1	$(GC_p) = -1.8$	for $A \leq 20 \text{ ft}^2$
	$(GC_p) = -1.4277 - 0.2861 \log A$	for $20 \leq A \leq 100 \text{ ft}^2$
	$(GC_p) = -2.0$	for $A \geq 100 \text{ ft}^2$
Zones 2r	$(GC_p) = -2.9$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -3.3612 + 0.4612 \log A$	for $10 \leq A \leq 200 \text{ ft}^2$
	$(GC_p) = -2.3$	for $A \geq 200 \text{ ft}^2$
Zones 2e	$(GC_p) = -2.3$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -2.4537 + 0.1537 \log A$	for $10 \leq A \leq 200 \text{ ft}^2$
	$(GC_p) = -2.1$	for $A \geq 200 \text{ ft}^2$
Zone 3	$(GC_p) = -2.9$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -3.8992 + 0.9992 \log A$	for $10 \leq A \leq 200 \text{ ft}^2$
	$(GC_p) = -1.6$	for $A \geq 200 \text{ ft}^2$

coefficients provided generally apply to facades with architectural features, such as balconies, ribs, and various facade textures.

Overhang pressures were determined by adding the effective uplift (GC_p) s implied in ASCE 7-10. These effective uplift

Table C30.3-8. Hip Roofs, $20^\circ < \theta \leq 27^\circ$ (Figure 30.3-2G)

Positive $h/B \geq 0.8$		
All Zones	$(GC_p) = 0.7$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = 1.1000 - 0.4000 \log A$	for $10 \leq A \leq 100 \text{ ft}^2$
	$(GC_p) = 0.3$	for $A \geq 100 \text{ ft}^2$
Negative without overhang		
Zone 1	$(GC_p) = -1.4$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -2.0000 + 0.6000 \log A$	for $10 \leq A \leq 100 \text{ ft}^2$
	$(GC_p) = -0.8$	for $A \geq 100 \text{ ft}^2$
Zones 2e, 2r, and 3	$(GC_p) = -2.0$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -2.7686 + 0.7686 \log A$	for $10 \leq A \leq 200 \text{ ft}^2$
	$(GC_p) = -1.0$	for $A \geq 200 \text{ ft}^2$
Negative with overhang		
Zone 1	$(GC_p) = -1.9$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -2.0000 + 0.1000 \log A$	for $10 \leq A \leq 100 \text{ ft}^2$
	$(GC_p) = -1.8$	for $A \geq 100 \text{ ft}^2$
Zones 2e and 2r	$(GC_p) = -2.5$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -2.8843 + 0.3843 \log A$	for $10 \leq A \leq 200 \text{ ft}^2$
	$(GC_p) = -2.0$	for $A \geq 200 \text{ ft}^2$
Zone 3	$(GC_p) = -3.1$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -4.3298 + 1.2298 \log A$	for $10 \leq A \leq 200 \text{ ft}^2$
	$(GC_p) = -1.5$	for $A \geq 200 \text{ ft}^2$

(GC_p) s were computed by subtracting the (GC_p) values given for the roof (no overhang) case from the (GC_p) s given for the overhang case. The additional (GC_p) s are given in Fig. C30.3-1.

The following guidance is based on the collective judgment of the wind load committee. For “L-shaped,” “T-shaped,” and other “irregular” shapes, Fig. C30.3-2 depicts the roof and wall zones for use with Figs. 30.3-1, 30.3-2, 30.3-4, 30.3-5, 30.3-6, 30.4-1, and 30.5-1 for wind loads on components and cladding of buildings, showing the applicability to buildings that are rectangular in plan. To address buildings with nonrectangular plans, Fig. C30.3-2 can be used for guidance in applying the requirements. When an outward corner protrudes less than the distance a from the wall, neither Zone 3 nor 5 are required; however, when the outward protrusion is greater than a , Zones 3 and 5 are required. Reentrant (interior) corners do not require Zones 3 or 5. For corners that have an included interior angle greater than 135° , neither Zone 3 nor 5 is required. To determine the length of a , a rectangle which enclosed the building is drawn over the building plan. The dimensions of this rectangle are used to determine the horizontal dimensions for the calculation of a .

Figs. 30.3-4, 30.3-5A, and 30.3-5B. These figures present values of (GC_p) for the design of roof C&C for buildings with

Table C30.3-9. Hip Roofs, $27^\circ < \theta \leq 45^\circ$, No Overhang (Figure 30.3-2H)

Positive		
All Zones	$(GC_p) = 0.9$	for $A \leq 2 \text{ ft}^2$
	$(GC_p) = 1.0063 - 0.3532 \log A$	for $2 \leq A \leq 100 \text{ ft}^2$
	$(GC_p) = 0.3$	for $A \geq 100 \text{ ft}^2$
Negative		
Zone 1	$(GC_p) = -0.6175 - 0.02000$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -1.0191 - 0.02500 + [0.4016 + 0.00500] \log A$	for $10 \leq A \leq 200 \text{ ft}^2$
	$(GC_p) = -0.0950 - 0.01350$	for $A \geq 200 \text{ ft}^2$
Zone 2e	$(GC_p) = 0.2000 - 0.06700$	for $A \leq 2 \text{ ft}^2$
	$(GC_p) = -0.8000 + \left[\frac{\log(280 - 50)(0.06700 - 1)}{0.301 - \log(280 - 50)} \right] + \left[\frac{1 - 0.06700}{0.3010 - \log(280 - 50)} \right] \log A$	for $2 \leq A \leq [280 - 50] \text{ ft}^2$
	$(GC_p) = -0.8$	for $A \geq [280 - 50] \text{ ft}^2$
Zones 2r	$(GC_p) = 1.0000 - 0.08200$	for $A \leq 5 \text{ ft}^2$
	$(GC_p) = 2.0746 - 0.12610 + [0.06300 - 1.5373] \log A$	for $5 \leq A \leq 100 \text{ ft}^2$
	$(GC_p) = -1.0000$	for $A \geq 100 \text{ ft}^2$
Zones 3	$(GC_p) = 1.2500 - 0.10800$	for $A \leq [9 - 0.13500] \text{ ft}^2$
	$(GC_p) = \left[\frac{0.18350 - 3.8230}{\log(9 - 0.13500) - 1.6990} \right] - 1.0 + \left[\frac{2.25 - 0.10800}{\log(9 - 0.13500) - 1.6990} \right] \log A$	for $[9 - 0.13500] \leq A \leq 50 \text{ ft}^2$
	$(GC_p) = -1.0000$	for $A \geq 50 \text{ ft}^2$

Table C30.3-10. Hip Roofs, $27^\circ < \theta \leq 45^\circ$, Overhang (Figure 30.3-2I)

Negative		
Zone 1	$(GC_p) = -1.4175 - 0.02000$	for $A \leq 10 \text{ ft}^2$
	$(GC_p) = -1.8191 - 0.02500 + [0.4016 + 0.00500] \log A$	for $10 \leq A \leq 200 \text{ ft}^2$
	$(GC_p) = -0.8950 - 0.01350$	for $A \geq 200 \text{ ft}^2$
Zone 2e	$(GC_p) = -0.6000 - 0.06700$	for $A \leq 2 \text{ ft}^2$
	$(GC_p) = -1.6000 + \left[\frac{\log(280 - 50)(0.06700 - 1)}{0.301 - \log(280 - 50)} \right] + \left[\frac{1 - 0.06700}{0.301 - \log(280 - 50)} \right] \log A$	for $2 \leq A \leq 280 - 50 \text{ ft}^2$
	$(GC_p) = -1.6000$	for $A \geq 280 - 50 \text{ ft}^2$
Zones 2r	$(GC_p) = 0.2000 - 0.08200$	for $A \leq 5 \text{ ft}^2$
	$(GC_p) = 1.2745 - 0.12610 + [0.06300 - 1.5373] \log A$	for $5 \leq A \leq 100 \text{ ft}^2$
	$(GC_p) = -1.8000$	for $A \geq 100 \text{ ft}^2$
Zones 3	$(GC_p) = 0.4500 - 0.10800$	for $A \leq 9 - 0.13500 \text{ ft}^2$
	$(GC_p) = \left[\frac{0.18350 - 3.823}{\log(9 - 0.1350) - 1.699} \right] - 1.8 + \left[\frac{2.25 - 0.1080}{\log(9 - 0.1350) - 1.699} \right] \log A$	for $9 - 0.13500 \leq A \leq 50 \text{ ft}^2$
	$(GC_p) = -1.8000$	for $A \geq 50 \text{ ft}^2$

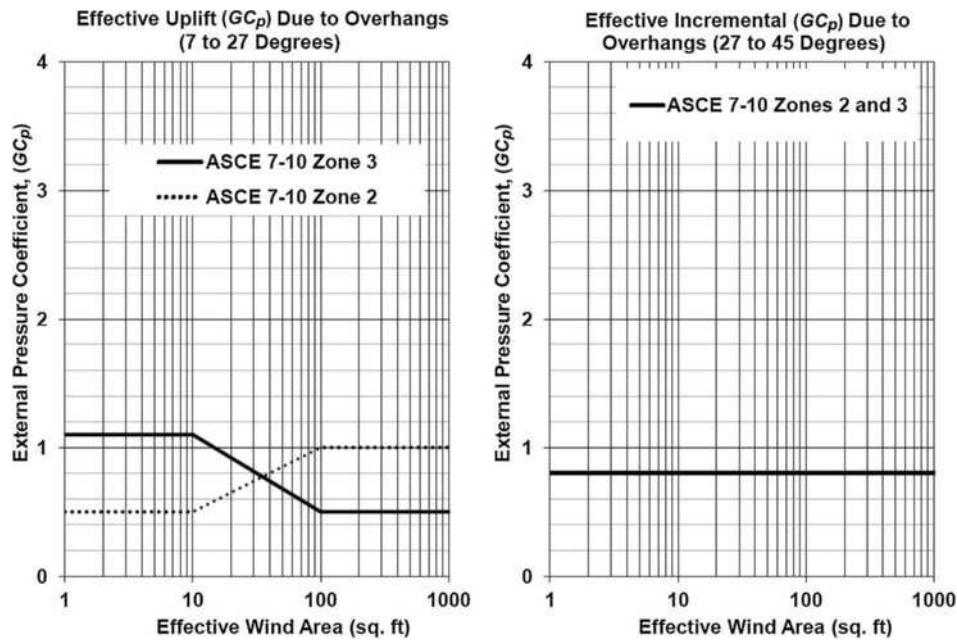


FIGURE C30.3-1 Effective Incremental (GC_p) Caused by Overhangs

multispan gable roofs and buildings with monoslope roofs. The coefficients are based on wind tunnel studies (Stathopoulos and Mohammadian 1986; Surry and Stathopoulos 1988; Stathopoulos and Saathoff 1991).

Fig. 30.3-6. The values of (GC_p) in this figure are for the design of roof C&C for buildings with sawtooth roofs and mean roof height, h , less than or equal to 60 ft (18.3 m). Note that the coefficients for corner zones on segment A differ from those coefficients for corner zones on the segments designated as B, C, and D. Also, when the roof angle is less than or equal to 10° , values of (GC_p) for regular gable roofs (Fig. 30.3-2A) are to be used. The coefficients included in Fig. 30.3-6 are based on wind tunnel studies reported by Saathoff and Stathopoulos (1992).

Fig. 30.3-7. This figure for cladding pressures on dome roofs is based on Taylor (1991). Negative pressures are to be applied to the entire surface because they apply along the full arc that is perpendicular to the wind direction and that passes through the top of the dome. Users are cautioned that only three shapes were available to define values in this figure: $h_D/D=0.5$, $f/D=0.5$; $h_D/D=0.0$, $f/D=0.5$; and $h_D/D=0.0$, $f/D=0.33$.

Fig. 30.5-1. The pressure coefficients shown in this figure reflect the results obtained from comprehensive wind tunnel studies carried out (Stathopoulos and Dumitrescu-Brulotte 1989). The availability of more-comprehensive wind tunnel data has also allowed a simplification of the zoning for pressure coefficients; flat roofs are now divided into three zones; and walls are represented by two zones.

The external pressure coefficients and zones given in Fig. 30.5-1 were established by wind tunnel tests on isolated “box-like” buildings (Akins and Cermak 1975; Peterka and Cermak 1975). Boundary-layer wind tunnel tests on high-rise buildings (mostly in downtown city centers) have shown that variations in pressure coefficients and the distribution of pressure on the different building facades are obtained (Templin and Cermak 1978). These variations are caused by building geometry, low attached buildings, nonrectangular cross sections, setbacks, and sloping surfaces. In addition, surrounding buildings contribute to the variations in pressure. Wind tunnel tests indicate that pressure

coefficients are not distributed symmetrically and can give rise to torsional wind loading on the building.

Boundary-layer wind tunnel tests that include modeling of surrounding buildings permit the establishment of more exact magnitudes and distributions of (GC_p) for buildings that are not isolated or “boxlike” in shape.

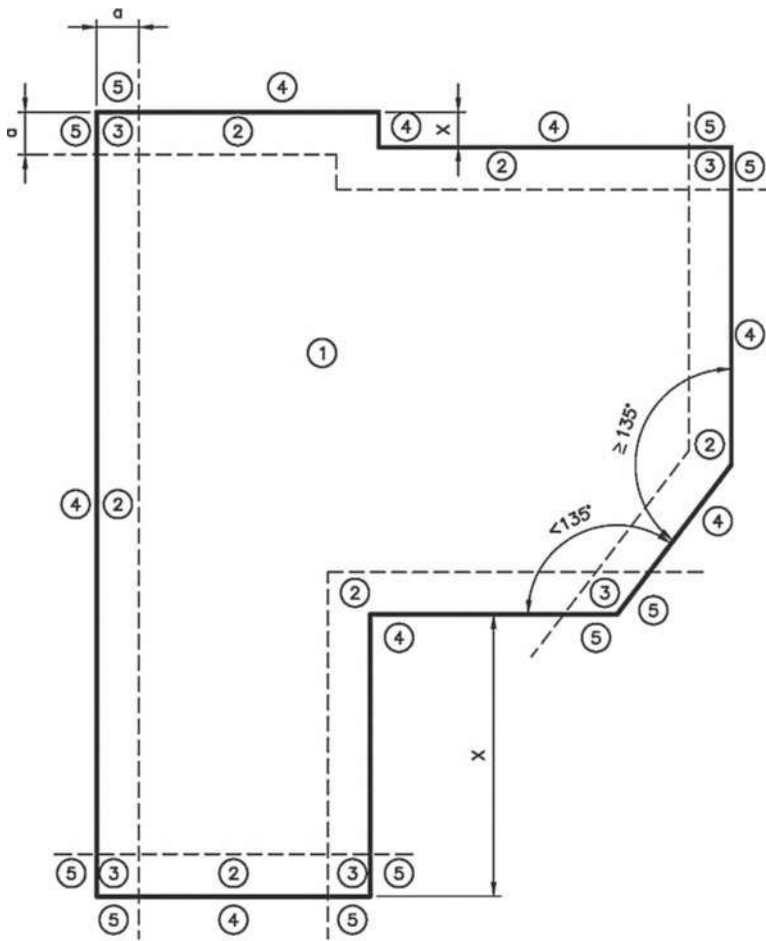
PART 1: LOW-RISE BUILDINGS

The C&C tables in Fig. 30.4-1 are a tabulation of the pressures on an enclosed, regular, 30-ft- (9.1-m)-high building with a roof as described. The pressures can be modified to a different exposure and height with the same adjustment factors as the MWFRS pressures. For the designer to use Part 2 for the design of the C&C, the building must conform to all five requirements in Section 30.4; otherwise, one of the other procedures specified in Section 30.1.1 must be used.

PART 3: BUILDINGS WITH $h > 60$ ft ($h > 18.3$ m)

In Eq. (30.5-1) a velocity pressure term, q_i , appears that is defined as the “velocity pressure for internal pressure determination.” The positive internal pressure is dictated by the positive exterior pressure on the windward face at the point where there is an opening. The positive exterior pressure at the opening is governed by the value of q at the level of the opening, not q_h . For positive internal pressure evaluation, q_i may conservatively be evaluated at height h ($q_i = q_h$). For low buildings, this height does not make much difference, but for the example of a 300-ft- (91.4-m)-tall building in Exposure B with the highest opening at 60 ft (18.3 m), the difference between q_{300} and q_{60} represents a 59% increase in internal pressure. This increase is unrealistic and represents an unnecessary degree of conservatism. Accordingly, $q_i = q_z$ for positive internal pressure evaluation in partially enclosed buildings where height z is defined as the level of the highest opening in the building that could affect the positive internal pressure. For buildings sited in wind-borne debris regions, glazing that is not impact-resistant or protected with an impact-protective system, q_i should be treated as an opening.

Diagram



Notation

a = 10% of least horizontal dimension or $0.4h$, whichever is smaller, but not less than either 4% of least horizontal dimension or 3 ft (0.9 m).

X = Offset dimension, in ft (m).

Notes

1. Labels marked on the roof plan indicate roof zones; labels marked outside the roof plan indicate wall zones.
2. If $X \leq a$, then Zone 3 and Zone 5 need not be applied at that corner.
3. If the interior angle is $\geq 135^\circ$, then Zone 3 and Zone 5 need not be applied at that corner.

FIGURE C30.3-2 Plan View of Roof and Wall Zones for Component and Cladding Loads on Buildings with Nonrectangular Plans

**PART 4: BUILDINGS WITH $60 \text{ ft} < h \leq 160 \text{ ft}$
($18.3 \text{ m} < h \leq 48.8 \text{ m}$) (SIMPLIFIED)**

C30.6 BUILDING TYPES

This section was added to ASCE 7-10 to cover the common practical case of enclosed buildings up to height $h = 160 \text{ ft}$ ($h = 48.8 \text{ m}$). Table 30.6-2 includes wall and roof pressures for flat roofs ($\theta \leq 7^\circ$), gable roofs, hip roofs, monoslope roofs, and mansard roofs. Pressures are derived from Fig. 30.5-1 (flat roofs) for $h > 60 \text{ ft}$ ($h > 18.3 \text{ m}$). For flat, gable, hip, monoslope, and mansard roofs with $h \leq 60 \text{ ft}$ ($h \leq 18.3 \text{ m}$) and all roof slopes,

reference is made to the roof and wall pressures tabulated in Fig. 30.4-1. For hip and gable roofs with $h > 60 \text{ ft}$ ($h > 18.3 \text{ m}$), reference is made to Fig. 30.5-1, Note 6, which permits the use of roof pressure coefficients in Figs. 30.3-2A through 30.3-2I as defined for low-rise buildings for these roof shapes if the appropriate q_h is used. Similarly, the roof pressure coefficients in Fig. 30.3-5A and B for monoslope roofs from Part 3 are permitted. Pressures were selected for each zone that encompasses the largest pressure coefficients for the comparable zones from the different roof shapes. Thus, for some cases, the pressures tabulated are conservative in order to maintain simplicity. The (GC_p) values from these figures were combined

with an internal pressure coefficient (± 0.18) to obtain a net coefficient from which pressures were calculated. The tabulated pressures are applicable to the entire zone shown in the various figures.

Pressures in Table 30.6-2 are shown for an effective wind area of 10 ft² (0.93 m²). A reduction factor is also shown to obtain pressures for larger effective wind areas. The reduction factors are based on the graph of external pressure coefficients shown in the figures in Part 3 and are based on the most conservative reduction for each zone from the various figures.

Note that the roof pressures tabulated for buildings with $h \leq 60$ ft ($h \leq 18.3$ m) in Fig. 30.4-1 are based on $h = 30$ ft ($h = 9.1$ m) and Exposure B. An adjustment factor λ is applied to the tabulated pressures for other heights and exposures. The tabulated pressures in Table 30.6-2 are based on Exposure C. An adjustment factor (EAF) from Eq. (30.6-1) is applied for other exposures.

C30.6.1 Wind Load: Components and Cladding.

C30.6.1.2 Parapets. Parapet C&C wind pressures can be obtained from the tables as shown in the parapet figures. The pressures obtained are slightly conservative based on the net pressure coefficients for parapets compared to roof zones from Part 3. Two load cases must be considered based on pressures applied to both windward and leeward parapet surfaces, as shown in Fig. 30.6-1.

C30.6.1.3 Roof Overhangs. C&C pressures for roof overhangs for flat and monoslope roofs with $h > 60$ ft ($h > 18.3$ m) can be obtained from the tables as shown in Fig. 30.6-2. These pressures are slightly conservative and are based on the external pressure coefficients from Part 3. Pressures for roof overhangs in flat, gable, hip, monoslope, and mansard roofs with $h \leq 60$ ft ($h \leq 18.3$ m) can be obtained directly from the tabulated pressures in Fig. 30.4-1.

PART 5: OPEN BUILDINGS

C30.7 BUILDING TYPES

In determining loads on C&C elements for open building roofs using Figs. 30.7-1, 30.7-2, and 30.7-3, it is important for the designer to note that the net pressure coefficient, C_N , is based on contributions from the top and bottom surfaces of the roof. This fact implies that the element receives load from both surfaces. Such would not be the case if the surface below the roof were separated structurally from the top roof surface. In this case, the pressure coefficient should be separated for the effect of top and bottom pressures, or conservatively, each surface could be designed using the C_N value from Figs. 30.7-1, 30.7-2, and 30.7-3.

PART 7: NONBUILDING STRUCTURES

C30.12 CIRCULAR BINS, SILOS, AND TANKS WITH $h \leq 120$ ft ($h \leq 36.5$ m)

Section 30.12 contains the provisions for determining wind pressures on silo and tank walls and roofs. The results of isolated and grouped silos are largely based on Australian Standards (Standards Australia 2011), and the wind tunnel tests by Sabransky and Melbourne (1987) and Macdonald et al. (1988, 1990). Significant increases in the mean pressures of grouped silos were found in the wind tunnel tests, so the provisions of grouped tanks and silos are specified in this section.

Table C30.12-1 Mean Pressure Coefficients ($GC_p - GC_{pi}$) for Open-Topped Tanks

Angle α	Aspect Ratio H/D					
	0.25	0.50	1	2	3	4
0°	1.69	1.80	1.9	2	2.07	2.11
15°	1.39	1.50	1.6	1.7	1.77	1.81
30°	0.99	1.10	1.2	1.3	1.37	1.41
45°	0.39	0.50	0.6	0.7	0.77	0.81
60°	-0.01	-0.01	-0.1	-0.1	-0.13	-0.09
75°	-0.11	-0.31	-0.5	-0.7	-0.83	-0.89
90°	-0.11	-0.31	-0.5	-0.7	-0.83	-0.89
105°	-0.01	-0.11	-0.2	-0.3	-0.33	-0.29
120°	0.09	0.10	0.2	0.2	0.27	0.21
135°	0.29	0.30	0.4	0.5	0.47	0.51
150°	0.29	0.40	0.5	0.5	0.57	0.61
165°	0.29	0.40	0.5	0.5	0.57	0.61
180°	0.29	0.40	0.5	0.5	0.57	0.61

Note: D = diameter of circular structure, in ft (m); H = height in ft (m); α = angle from the wind direction to a point on the wall of a circular bin, silo, or tank, in degrees.

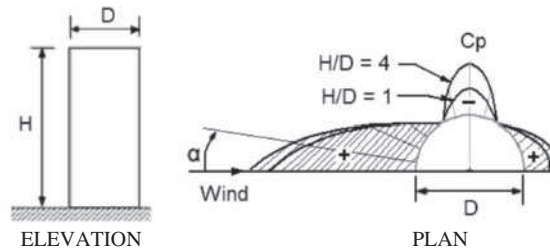
C30.12.2 External Walls of Isolated Circular Bins, Silos, and Tanks. This section specifies the external pressure coefficients ($GC_{p(\alpha)}$) for the walls of circular bins, silos, and tanks. The pressure coefficients for isolated silos are adopted from Australian Standards (Standards Australia 2011).

C30.12.3 Internal Surface of Exterior Walls of Isolated Open-Topped Circular Bins, Silos, and Tanks. This section specifies the internal pressure coefficients (GC_{pi}) for the walls of circular bins, silos, and tanks. The internal pressure coefficients (GC_{pi}) are adopted from Standards Australia (2011). Based on the wind tunnel test results, mean pressures on walls for open-topped bins, silos, and tanks are different from the values of circular bins, silos, and tanks with flat or conical roofs. Table C30.12-1 lists the mean pressure coefficients ($GC_p - GC_{pi}$) for open-topped circular bins, silos, and tanks, based on Eqs. (30.12-2) and (30.12-5). The distribution of the external pressure around the perimeter of the wall is shown in Fig. C30.12-1.

C30.12.4 Roofs of Isolated Circular Bins, Silos, and Tanks. This section specifies the external pressure coefficients (GC_p) for the roofs of circular bins, silos, and tanks. Two conditions are covered as shown in Fig. 30.12-2: Class 1 roofs have the roof angle $\theta < 10^\circ$, and Class 2 roofs have $10^\circ \leq \theta < 30^\circ$. Zone 1 pressures are defined differently that either increase with the increment of the silo heights for Class 1 roofs, or with the silo or tank diameters for Class 2 roofs. For cladding design, Zone 3 pressures are specified for the local pressures near the windward edges applicable to all classes, and Zone 4 is specified for the region near the cone apex used for Class 2b roofs only. Fig. C30.12-2 is the graphic presentation of the elevation views for the external pressure coefficients (GC_p). For Class 1 roofs, the external pressure coefficients are based on comparisons of domed roofs and flat roofs from Chapter 27 of ASCE 7-10 for maximum uplift conditions. The results of Class 2 roofs are consistent with data of Sabransky and Melbourne (1987) and Macdonald et al. (1988).

C30.12.6 Roofs and Walls of Grouped Circular Bins, Silos, and Tanks. For grouped silos, (GC_p) values for roofs and walls are largely based on AS/ NZS 1170.2 (Standards Australia 2011) and wind tunnel tests by Sabransky and Melbourne (1987) and

Diagrams



Notation

C_p = External pressure coefficient to be used in determination of wind loads for buildings.

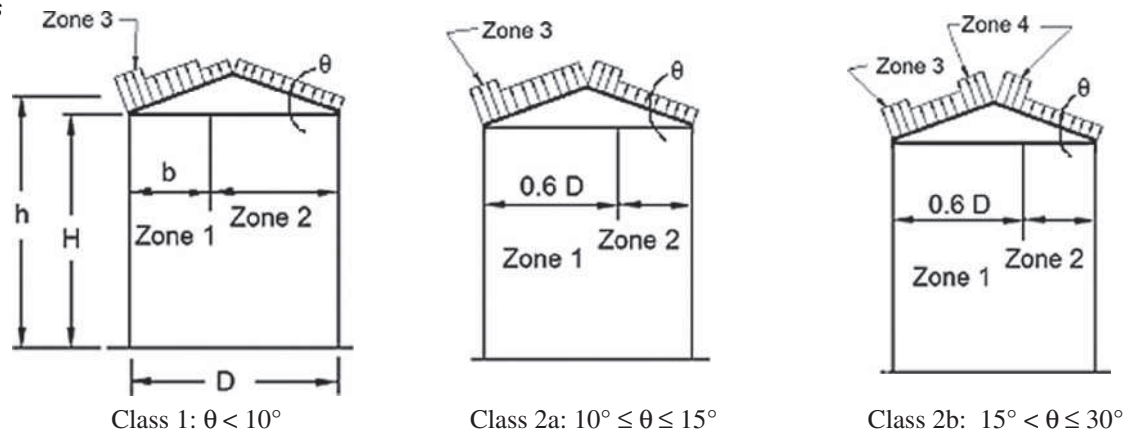
D = Diameter of a circular structure, in ft (m).

H = Height of solid cylinder, in ft (m).

α = Angle from the wind direction to a point on the wall of a circular bin, silo, or tank, in degrees (see Section 30.12.2).

FIGURE C30.12-1 Mean Pressure Coefficients ($(GC_p)-(GC_{pi})$) for Open-Topped Tanks

Diagrams



Notation

b = Horizontal dimension specified for Zone 1 of a conical roof, in ft (m). For roof angles less than 10 degrees, b is calculated from the table of external pressure coefficients in Fig. 30.12-2 (e.g., $b = 0.5D$ for $H/D = 0.5$.) For roof angles equal to or larger than 10 degrees, $b = 0.6D$. (So for Class 2a and 2b, $b = 0.6D$.)

D = Diameter of a circular structure, in ft (m).

h = Mean roof height, in ft (m).

H = Height of the solid cylinder, in ft (m).

θ = Angle of plane of roof from horizontal in degrees.

FIGURE C30.12-2 External Pressure Coefficients (GC_p) for Roofs

Macdonald et al. (1990). Test results of an in-line group of three silos with a clear spacing of $0.25D$ between nearest adjacent walls ($1.25D$ center-to-center) by Sabransky and Melbourne (1987) indicated that the mean pressure coefficient between the gaps increased by 70% compared to the one for the isolated silo. A similar result was observed for the roof near the wall of the silo. It was concluded that a clear spacing of $0.25D$ produced the maximum interference between two finite cylinders.

Test results of an in-line group of five silos with various center-to-center spacings by Macdonald et al. (1990) indicated that the region of positive pressure on the windward side spans a larger angular sector of the circumference than that for an isolated silo, and high negative mean pressures occur near the point of

shortest distance between the adjacent silos and at the outside corners of the groups.

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